

Spectral Admissibility and Non-Abelian Mode Resilience in the Q_8 Cayley Graph under Bounded Flux

Working note – Cosmochrony programme

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Abstract

We establish the first analytically tractable instance of the spectral admissibility programme outlined in the Cosmochrony framework. On the undirected Cayley graph of the quaternion group Q_8 , with generators $S = \{\pm i, \pm j, \pm k\}$, the local bounded-flux constraint $|\partial_t \chi_v| \leq c_\chi$ induces a hierarchy of maximal admissible effective amplitudes across irreducible spectral sectors.

In the linearised dispersion regime, the non-abelian irreducible sector admits a maximal effective amplitude larger than the nontrivial abelian sectors by a factor of $\sqrt{4/3} \approx 1.155$. Resolving explicitly the basis dependence in the degenerate ρ_4 eigenspace, we show that this advantage holds for the physically relevant effective amplitude $A_n = a_n \|\psi_n\|_\infty$, but not for the raw amplitude a_n .

We then perform the weakly non-linear modal reduction of the discrete Dirac–Born–Infeld (DBI) Lagrangian and derive the first amplitude-dependent correction to the dispersion relation, governed by the quartic shape factor $\kappa_n = \sum_v \psi_n(v)^4$. When expressed in terms of the effective amplitude, this correction depends only on the Laplacian eigenvalue λ_n , leaving the spectral hierarchy unchanged.

Finally, solving the one-mode DBI dynamics exactly on the Q_8 Cayley graph, we show that the admissibility envelope

$$A_n^{\max} = \frac{c_\chi}{\sqrt{\lambda_n}}$$

coincides with the exact saturation boundary of the DBI dynamics. The spectral hierarchy therefore remains structurally stable beyond the perturbative regime.

This provides a minimal explicit demonstration that bounded substrate capacity can favour non-abelian relational modes purely through spectral admissibility, without invoking additional dynamical selection principles.

We further extend the analysis to the binary icosahedral group $2I$, the largest finite subgroup of $SU(2)$. Using the order-10 generating classes, the same admissibility envelope produces a four-level spectral hierarchy across the spin sectors of $2I$, with an admissibility contrast $\sqrt{14/9}$ between the most and least admissible non-trivial sectors. This provides the second explicit instance of the spectral admissibility programme along the chain

$$Q_8 \subset 2I \subset SU(2).$$

More broadly, it suggests that non-abelian symmetry sectors may emerge as the dynamically most resilient components of the relational spectrum under bounded-flux constraints.

1 Setting and Motivation

The Cosmochrony framework identifies the physical spectrum with the residue of two successive constraints acting on the relational operator $\mathcal{L}_\chi$:

1. **Topological discretisation:** the fibre structure of the non-injective projection Π organises modes into discrete winding sectors.

2. **Bounded-flux filtering:** the saturation bound $|\partial_t \chi| \leq c_\chi$ eliminates modes whose local evolution would exceed the substrate's relational capacity.

The goal of this note is to make step 2 explicit and calculable on the simplest non-trivial finite graph encoding a quaternionic algebraic structure, as a minimal discrete proxy for $SU(2)$ -type sectors.

2 The Q_8 Cayley Graph

Let $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ denote the quaternion group of order 8. We define the undirected Cayley graph $\Gamma(Q_8, S)$ with generating set

$$S = \{\pm i, \pm j, \pm k\}. \quad (1)$$

Since S is closed under inversion ($s^{-1} \in S$ for all $s \in S$) and does not contain the identity, $\Gamma(Q_8, S)$ is a well-defined, connected, undirected, 6-regular graph on 8 vertices.

Remark 1. *The choice $S = \{i, j, k\}$ without inverses would yield a directed graph with an asymmetric adjacency operator. For a self-adjoint Laplacian, the full symmetric set $S = \{\pm i, \pm j, \pm k\}$ is required. The resulting graph is 6-regular, not 3-regular.*

2.1 Adjacency Operator and Graph Laplacian

The adjacency operator A of $\Gamma(Q_8, S)$ acts on functions $f: Q_8 \rightarrow \mathbb{C}$ by

$$(Af)(g) = \sum_{s \in S} f(gs) = \sum_{s \in \{\pm i, \pm j, \pm k\}} f(gs). \quad (2)$$

The combinatorial graph Laplacian is

$$\mathcal{L} = dI - A, \quad d = |S| = 6, \quad (3)$$

where I is the identity on $\ell^2(Q_8)$. The eigenvalues of $\mathcal{L}$ are $\lambda = 6 - \mu$, where μ ranges over the eigenvalues of A .

3 Spectral Decomposition via Representation Theory

Since Q_8 is a finite group, $\ell^2(Q_8) \cong \mathbb{C}[Q_8]$ decomposes as a Q_8 -module into irreducible representations. By the Peter–Weyl theorem for finite groups:

$$\mathbb{C}[Q_8] \cong \bigoplus_{\rho \in \widehat{Q_8}} V_\rho^{\oplus \dim \rho}, \quad (4)$$

where the sum runs over the five irreducible representations of Q_8 .

3.1 Irreducible Representations of Q_8

Q_8 has five irreducible representations:

Label	Dimension	$\rho(i)$	$\rho(j)$	$\rho(k)$
ρ_0 (trivial)	1	1	1	1
ρ_1	1	1	-1	-1
ρ_2	1	-1	1	-1
ρ_3	1	-1	-1	1
ρ_4 (quaternionic)	2	$\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$

The dimension check: $1^2 + 1^2 + 1^2 + 1^2 + 2^2 = 8 = |Q_8|$.

3.2 Eigenvalues of the Adjacency Operator

For any irreducible representation ρ , the adjacency operator acts as the scalar $\mu_\rho \cdot I_{V_\rho}$ on V_ρ . Since S is a union of conjugacy classes of Q_8 , the corresponding element of the group algebra is central, and Schur's lemma gives

$$\mu_\rho = \frac{1}{\dim \rho} \sum_{s \in S} \chi_\rho(s), \quad (5)$$

where χ_ρ is the character of ρ .

1D sectors. For the trivial representation ρ_0 , $\chi_{\rho_0}(s) = 1$ for all $s \in S$, so $\sum_{s \in S} \chi_{\rho_0}(s) = 6$ and

$$\mu_{\rho_0} = 6, \quad \lambda_{\rho_0} = 6 - 6 = 0. \quad (6)$$

For ρ_1 , $\chi_{\rho_1}(\pm i) = 1$ and $\chi_{\rho_1}(\pm j) = \chi_{\rho_1}(\pm k) = -1$, so $\sum_{s \in S} \chi_{\rho_1}(s) = 2(1) + 4(-1) = -2$, giving $\mu_{\rho_1} = -2$ and $\lambda_{\rho_1} = 8$. By the symmetry of Q_8 under permutations of $\{i, j, k\}$, $\mu_{\rho_2} = \mu_{\rho_3} = -2$ and $\lambda_{\rho_2} = \lambda_{\rho_3} = 8$.

2D sector. For the quaternionic representation ρ_4 , we compute the character at each generator. The matrices $\rho_4(i)$, $\rho_4(-i)$, $\rho_4(j)$, $\rho_4(-j)$, $\rho_4(k)$, $\rho_4(-k)$ each have trace zero. Thus $\sum_{s \in S} \chi_{\rho_4}(s) = 0$ and

$$\mu_{\rho_4} = 0, \quad \lambda_{\rho_4} = 6. \quad (7)$$

3.3 Summary of the Spectrum

Sector	λ_n	Multiplicity	Type
ρ_0	0	1	Trivial zero mode
ρ_1, ρ_2, ρ_3	8	3	Nontrivial abelian
ρ_4	6	4	Non-abelian quaternionic

Total dimension: $1 + 3 + 4 = 8$.

4 Bounded-Flux Admissibility Condition

4.1 Setup

Let the substrate field $\chi: Q_8 \times \mathbb{R} \rightarrow \mathbb{R}$ evolve on the graph, subject to the local bounded-flux constraint

$$|\partial_t \chi_v(t)| \leq c_\chi \quad \forall v \in Q_8, \forall t. \quad (8)$$

Consider a single-mode excitation

$$\chi_v(t) = a_n \psi_n(v) \cos(\omega_n t), \quad (9)$$

where ψ_n is an eigenmode of $\mathcal{L}$ with eigenvalue λ_n , normalised in $\ell^2(Q_8)$, and $a_n > 0$ is the mode amplitude. The time derivative at vertex v is

$$\partial_t \chi_v(t) = -a_n \omega_n \psi_n(v) \sin(\omega_n t). \quad (10)$$

The constraint (8) requires

$$|\omega_n| \cdot a_n \cdot |\psi_n(v)| \leq c_\chi \quad \forall v \in Q_8. \quad (11)$$

The binding constraint is at the vertex where $|\psi_n(v)|$ is maximal:

$$|\omega_n| \cdot a_n \cdot \|\psi_n\|_\infty \leq c_\chi. \quad (12)$$

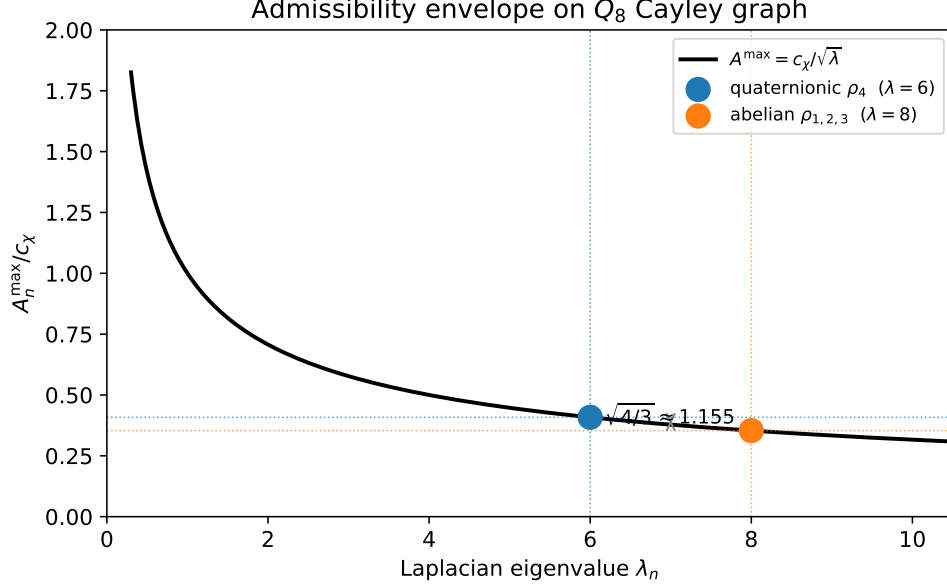


Figure 1: Admissibility envelope on the Q_8 Cayley graph with $S = \{\pm i, \pm j, \pm k\}$. The maximal effective amplitude $A_n^{\max} = c_\chi / \sqrt{\lambda_n}$ is larger in the quaternionic non-abelian sector ρ_4 ($\lambda = 6$) than in the non-trivial abelian sectors ($\lambda = 8$), by the factor $\sqrt{4/3}$.

4.2 Effective Amplitude and Admissibility Envelope

Definition 1. The effective amplitude of a mode (ψ_n, a_n) is

$$A_n := a_n \|\psi_n\|_\infty. \quad (13)$$

A mode is dynamically admissible in the linearised regime if

$$|\omega_n| \cdot A_n \leq c_\chi. \quad (14)$$

In the linearised dispersion approximation, $\omega_n^2 \approx \lambda_n$, so the maximal admissible effective amplitude is

$$A_n^{\max} = \frac{c_\chi}{\sqrt{\lambda_n}}. \quad (15)$$

This defines the admissibility envelope in the (λ_n, A_n) plane.

Remark 2. The linearised dispersion $\omega_n^2 = \lambda_n$ is valid for $A_n \ll c_\chi / \sqrt{\lambda_n}$. The non-linear Born-Infeld correction modifies this relation near saturation and requires a modal reduction of the DBI action. The envelope (15) should therefore be understood first as the admissibility boundary in the weakly non-linear regime.

4.3 Explicit Eigenmodes in the ρ_4 Sector

The non-abelian sector ρ_4 has multiplicity 4 in $\ell^2(Q_8)$, so the quantity $\|\psi_n\|_\infty$ is not determined until an explicit basis of the eigenspace is chosen. A natural orthonormal basis is provided by the matrix coefficients of the quaternionic irreducible representation.

For a finite group G and an irreducible representation ρ , the standard Peter-Weyl basis of the corresponding isotypic component is given by

$$\psi_{uv}(g) = \sqrt{\frac{\dim \rho}{|G|}} \rho(g)_{uv}, \quad u, v \in \{1, \dots, \dim \rho\}. \quad (16)$$

For Q_8 and ρ_4 , this becomes

$$\psi_{uv}(g) = \frac{1}{2} \rho_4(g)_{uv}, \quad u, v \in \{1, 2\}, \quad (17)$$

since $\dim \rho_4 = 2$ and $|Q_8| = 8$.

Using the explicit matrices

$$\rho_4(1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \rho_4(-1) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (18)$$

$$\rho_4(i) = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad \rho_4(j) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \rho_4(k) = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \quad (19)$$

one finds that the diagonal modes ψ_{11}, ψ_{22} are supported on $\{\pm 1, \pm i\}$, while the off-diagonal modes ψ_{12}, ψ_{21} are supported on $\{\pm j, \pm k\}$. In all four cases,

$$|\psi_{uv}(g)| \in \left\{0, \frac{1}{2}\right\}, \quad (20)$$

hence

$$\|\psi_{uv}\|_\infty = \frac{1}{2}, \quad u, v \in \{1, 2\}. \quad (21)$$

For the one-dimensional nontrivial abelian sectors ρ_1, ρ_2, ρ_3 , the normalised eigenmodes are the corresponding characters scaled by $|Q_8|^{-1/2}$. Since these characters take values in $\{\pm 1\}$, one has

$$\|\psi_{\rho_k}\|_\infty = \frac{1}{\sqrt{8}}, \quad k = 1, 2, 3. \quad (22)$$

4.4 Basis-Resolved Admissibility

Proposition 1 (Basis-resolved admissibility on Q_8). *In the canonical Peter–Weyl basis of $\ell^2(Q_8)$, the maximal admissible effective amplitudes satisfy*

$$A_{\rho_4}^{\max} = \frac{c_\chi}{\sqrt{6}} > \frac{c_\chi}{\sqrt{8}} = A_{\rho_k}^{\max}, \quad k = 1, 2, 3, \quad (23)$$

whereas the corresponding maximal raw amplitudes satisfy

$$a_{\rho_4}^{\max} = \frac{c_\chi}{\sqrt{6} \|\psi_{\rho_4}\|_\infty} = \frac{2}{\sqrt{6}} c_\chi \approx 0.816 c_\chi, \quad (24)$$

$$a_{\rho_k}^{\max} = \frac{c_\chi}{\sqrt{8} \|\psi_{\rho_k}\|_\infty} = c_\chi, \quad k = 1, 2, 3. \quad (25)$$

Thus the non-abelian sector is favoured in effective amplitude, but not in raw amplitude.

Proof. The effective-amplitude statement follows from $A_n^{\max} = c_\chi / \sqrt{\lambda_n}$ together with $\lambda_{\rho_4} = 6$ and $\lambda_{\rho_k} = 8$. The raw-amplitude statement follows from $a_n^{\max} = A_n^{\max} / \|\psi_n\|_\infty$, using $\|\psi_{\rho_4}\|_\infty = 1/2$ and $\|\psi_{\rho_k}\|_\infty = 1/\sqrt{8}$. $\square$

4.5 Weakly Non-linear Modal DBI Reduction

To incorporate the first Born–Infeld correction, consider the discrete DBI Lagrangian

$$L = -c_\chi^2 \sum_v \sqrt{1 - \dot{\chi}_v^2 / c_\chi^2} - \frac{1}{2} \sum_{v,w} \chi_v \mathcal{L}_{vw} \chi_w. \quad (26)$$

For a single eigenmode,

$$\chi_v(t) = q(t)\psi_n(v), \quad (27)$$

with $\mathcal{L}\psi_n = \lambda_n\psi_n$ and $\|\psi_n\|_2 = 1$, this reduces to the exact one-mode Lagrangian

$$L_n[q] = -c_\chi^2 \sum_v \sqrt{1 - \psi_n(v)^2 \dot{q}^2 / c_\chi^2} - \frac{1}{2} \lambda_n q^2. \quad (28)$$

Expanding the square root for $|\dot{q}| \ll c_\chi$ gives

$$L_n[q] = \text{const} + \frac{1}{2} \dot{q}^2 + \frac{\kappa_n}{8c_\chi^2} \dot{q}^4 - \frac{1}{2} \lambda_n q^2 + \mathcal{O}(c_\chi^{-4}), \quad (29)$$

where

$$\kappa_n := \sum_v \psi_n(v)^4. \quad (30)$$

The corresponding Euler–Lagrange equation is

$$\left(1 + \frac{3\kappa_n}{2c_\chi^2} \dot{q}^2\right) \ddot{q} + \lambda_n q = 0. \quad (31)$$

Using the harmonic-balance ansatz

$$q(t) \approx a_n \cos(\omega_n t), \quad (32)$$

and keeping only the fundamental harmonic, one obtains the first amplitude-dependent correction

$$\omega_n^2(a_n) \approx \lambda_n \left(1 - \frac{3\kappa_n \lambda_n a_n^2}{8c_\chi^2}\right), \quad (33)$$

or equivalently

$$\omega_n(a_n) \approx \sqrt{\lambda_n} \left(1 - \frac{3\kappa_n \lambda_n a_n^2}{16c_\chi^2}\right). \quad (34)$$

The DBI correction is therefore a softening non-linearity.

In the canonical Q_8 basis, the quartic shape factor is explicit. For the one-dimensional nontrivial abelian sectors,

$$\kappa_{\rho_k} = \sum_{v \in Q_8} \left(\frac{1}{\sqrt{8}}\right)^4 = \frac{1}{8}, \quad k = 1, 2, 3. \quad (35)$$

For each canonical basis vector in the ρ_4 sector,

$$\kappa_{\rho_4} = 4 \left(\frac{1}{2}\right)^4 = \frac{1}{4}. \quad (36)$$

However, rewriting the correction in terms of the effective amplitude $A_n = a_n \|\psi_n\|_\infty$ gives

$$\omega_n^2(A_n) \approx \lambda_n \left(1 - \frac{3\lambda_n A_n^2}{8c_\chi^2} \frac{\kappa_n}{\|\psi_n\|_\infty^2}\right). \quad (37)$$

In the canonical Q_8 basis,

$$\frac{\kappa_{\rho_k}}{\|\psi_{\rho_k}\|_\infty^2} = \frac{(1/8)}{(1/\sqrt{8})^2} = 1, \quad \frac{\kappa_{\rho_4}}{\|\psi_{\rho_4}\|_\infty^2} = \frac{(1/4)}{(1/2)^2} = 1. \quad (38)$$

Therefore

$$\omega_n^2(A_n) \approx \lambda_n \left(1 - \frac{3\lambda_n A_n^2}{8c_\chi^2}\right) \quad (39)$$

in both the nontrivial abelian and non-abelian sectors.

At fixed effective amplitude, the non-abelian sector thus remains less penalised by the weakly non-linear DBI correction, solely because $\lambda_{\rho_4} = 6$ is smaller than $\lambda_{\rho_k} = 8$.

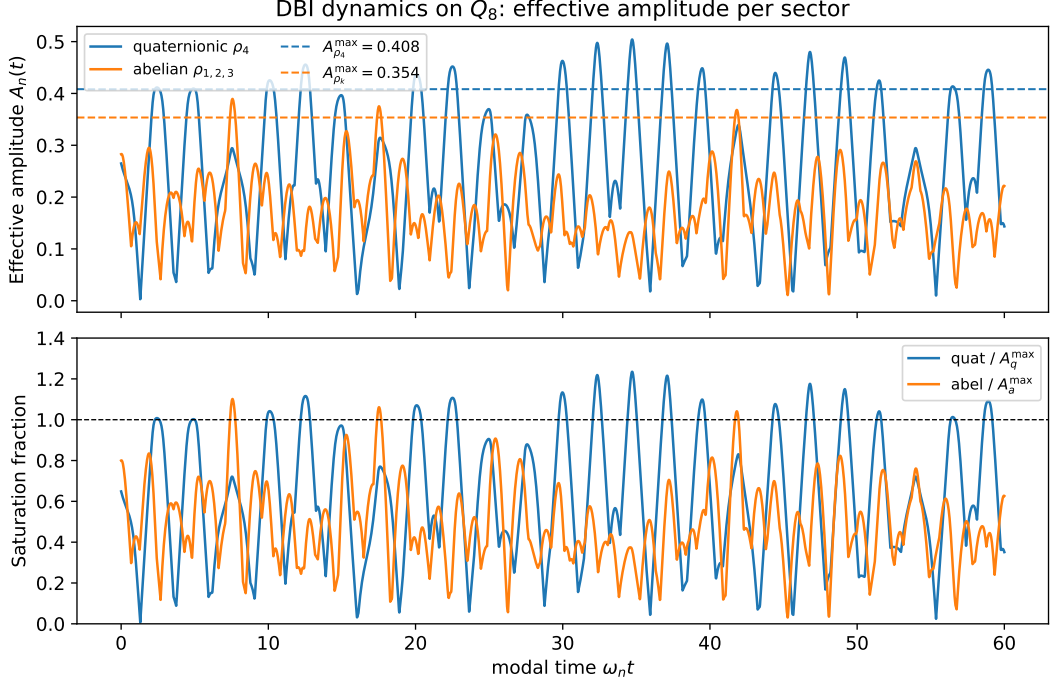


Figure 2: One-mode DBI dynamics on the Q_8 Cayley graph. Top: effective amplitudes in the quaternionic and abelian sectors. Bottom: corresponding saturation fractions relative to the exact admissibility bounds. The horizontal axis denotes the evolution parameter of the reduced modal dynamics. The quaternionic sector retains the largest admissible window, consistent with the spectral hierarchy derived from the bounded-flux constraint.

4.6 Non-perturbative DBI Envelope

The weakly non-linear reduction derived in Section 4.5 provides the leading amplitude correction to the dispersion relation. However, the discrete Dirac–Born–Infeld dynamics allows the single-mode problem to be solved exactly on the Q_8 Cayley graph.

Consider again a single eigenmode

$$\chi_v(t) = q(t)\psi_n(v),$$

with $\mathcal{L}\psi_n = \lambda_n\psi_n$ and $\|\psi_n\|_2 = 1$. Substituting this ansatz into the DBI Lagrangian Eq. (26) gives the exact one-mode Lagrangian

$$L_n[q] = -c_\chi^2 \sum_{v \in Q_8} \sqrt{1 - \frac{\psi_n(v)^2 \dot{q}^2}{c_\chi^2}} - \frac{1}{2} \lambda_n q^2. \quad (40)$$

Abelian sectors

For the nontrivial abelian representations ρ_1, ρ_2, ρ_3 , the normalized eigenmodes satisfy

$$|\psi_n(v)| = \frac{1}{\sqrt{8}} \quad \forall v \in Q_8.$$

The Lagrangian therefore becomes

$$L = -8c_\chi^2 \sqrt{1 - \frac{\dot{q}^2}{8c_\chi^2}} - \frac{1}{2} \lambda_n q^2. \quad (41)$$

Non-abelian sector

For the quaternionic sector ρ_4 in the canonical basis, the eigenmodes satisfy

$$|\psi_n(v)| = \begin{cases} \frac{1}{2} & \text{on four vertices,} \\ 0 & \text{on four vertices.} \end{cases}$$

The DBI sum therefore reduces to

$$L = -4c_\chi^2 \sqrt{1 - \frac{\dot{q}^2}{4c_\chi^2}} - 4c_\chi^2 - \frac{1}{2}\lambda_n q^2. \quad (42)$$

The constant term does not affect the dynamics, and the effective Lagrangian becomes

$$L = -4c_\chi^2 \sqrt{1 - \frac{\dot{q}^2}{4c_\chi^2}} - \frac{1}{2}\lambda_n q^2. \quad (43)$$

Energy conservation

Since the Lagrangian is time independent, the corresponding Hamiltonian is conserved. One obtains

$$E = 4c_\chi^2 \frac{1}{\sqrt{1 - \frac{\dot{q}^2}{4c_\chi^2}}} + \frac{1}{2}\lambda_n q^2. \quad (44)$$

At the turning points $\dot{q} = 0$, where $q = \pm a_n$, the energy becomes

$$E = 4c_\chi^2 + \frac{1}{2}\lambda_n a_n^2, \quad (45)$$

where a_n denotes the raw oscillation amplitude of the modal coordinate.

Velocity bound

The DBI square root imposes the kinematic bound

$$|\dot{q}| < 2c_\chi. \quad (46)$$

Expressed in terms of the field variable,

$$\chi_v = q\psi_n(v),$$

this corresponds exactly to the local flux constraint

$$|\partial_t \chi_v| \leq c_\chi.$$

Amplitude bound

The turning point occurs when the maximal vertex saturates the flux bound,

$$|\partial_t \chi_v| = |\dot{q}| |\psi_n(v)| = c_\chi.$$

Since $\|\psi_n\|_\infty$ is the maximal value of $|\psi_n(v)|$, this gives

$$a_n^{\max} = \frac{c_\chi}{\|\psi_n\|_\infty \sqrt{\lambda_n}}. \quad (47)$$

The corresponding effective amplitude

$$A_n = a_n \|\psi_n\|_\infty$$

therefore satisfies

$$A_n^{\max} = \frac{c_\chi}{\sqrt{\lambda_n}}. \quad (48)$$

Exact admissibility envelope

We thus obtain the non-perturbative result

$$A_n^{\max} = \frac{c_\chi}{\sqrt{\lambda_n}}, \quad (49)$$

which coincides exactly with the admissibility envelope derived from the linear bounded-flux argument.

The weakly non-linear correction therefore does not modify the location of the admissibility boundary. Instead, the DBI dynamics smoothly interpolates between the linear regime and a velocity-saturated regime as the amplitude approaches the bound.

This shows that the spectral admissibility envelope identified above is not merely a linear approximation but the exact saturation boundary for the single-mode DBI dynamics on the Q_8 Cayley graph.

5 Admissibility Hierarchy on the Q_8 Cayley Graph

Proposition 2 (Spectral admissibility hierarchy on Q_8). *In the undirected Cayley graph $\Gamma(Q_8, \{\pm i, \pm j, \pm k\})$, under the local bounded-flux constraint (8) and linearised dispersion, the maximal admissible effective amplitude satisfies*

$$A_{\rho_4}^{\max} = \frac{c_\chi}{\sqrt{6}} > \frac{c_\chi}{\sqrt{8}} = A_{\rho_k}^{\max}, \quad k = 1, 2, 3. \quad (50)$$

The non-abelian sector ρ_4 occupies a strictly larger admissible region in the space of effective mode amplitudes than any nontrivial abelian sector, by a factor

$$\frac{A_{\rho_4}^{\max}}{A_{\rho_k}^{\max}} = \sqrt{\frac{8}{6}} = \sqrt{\frac{4}{3}} \approx 1.155. \quad (51)$$

Proof. Direct substitution of $\lambda_{\rho_4} = 6$ and $\lambda_{\rho_k} = 8$ into the admissibility envelope (15). $\square$

Remark 3. *The trivial sector ρ_0 has $\lambda = 0$ and thus formally unbounded $A^{\max}$. It corresponds to the uniform zero mode of the connected graph. In the present analysis it is factored out, since we are interested only in non-trivial excitations above the connected background.*

Proposition 2, together with Proposition 1, establishes that, at fixed substrate capacity c_χ , the non-abelian irreducible sector of the quaternion Cayley graph sustains a larger local effective amplitude before violating the bounded-flux constraint, even though its raw amplitude bound is lower than that of the nontrivial abelian sectors.

In the language of the Cosmochrony programme:

- The admissibility envelope $A_n^{\max} = c_\chi/\sqrt{\lambda_n}$ defines a geometry of exclusion in the space of relational modes. Modes above the envelope are dynamically inadmissible.
- The non-abelian sector ρ_4 lies below this envelope for a larger range of effective amplitudes than the nontrivial abelian sectors ρ_1, ρ_2, ρ_3 . The bounded-flux constraint is therefore less restrictive for non-abelian modes, enlarging their admissible window by a factor of $\sqrt{4/3}$.
- This advantage is a statement about the physically relevant local variable $A_n = a_n \|\psi_n\|_\infty$, not about the raw amplitude a_n .
- In the canonical basis, the weakly non-linear DBI correction again favours the non-abelian sector through its smaller eigenvalue.

- The exact one-mode DBI analysis shows that the same envelope $A_n^{\max} = c_\chi / \sqrt{\lambda_n}$ remains the true saturation boundary.

Remark 4 (Real field and complex eigenmodes). *The substrate field χ is taken real-valued. The natural eigenmodes of the 2D sector are matrix coefficients of ρ_4 , which are complex-valued. One may either work in $\ell^2(Q_8, \mathbb{C})$ and take real and imaginary parts to obtain real eigenmodes, or directly choose a real basis for each eigenspace. The local admissibility criterion is intrinsically defined, but its expression in terms of (a_n, ψ_n) depends on the chosen basis within a degenerate eigenspace through $\|\psi_n\|_\infty$. This basis dependence is resolved explicitly in Section 4.3.*

6 Synthesis: Spectral Filtering by Bounded Flux

The analysis performed in this note identifies the effective amplitude

$$A_n = a_n \|\psi_n\|_\infty$$

as the physically relevant variable controlling the saturation of the substrate.

6.1 Local Flux Constraint as a Spectral Filter

The bounded-flux condition

$$|\partial_t \chi_v| \leq c_\chi$$

acts locally on each vertex of the relational substrate.

For a single eigenmode

$$\chi_v(t) = a_n \psi_n(v) \cos(\omega_n t),$$

the constraint becomes

$$|\omega_n| a_n \|\psi_n\|_\infty \leq c_\chi.$$

Introducing the effective amplitude

$$A_n = a_n \|\psi_n\|_\infty$$

yields the admissibility envelope

$$A_n^{\max} = \frac{c_\chi}{\sqrt{\lambda_n}}.$$

A key consequence is that localisation of the mode does not alter the admissibility bound itself. Localisation only determines how the raw amplitude a_n translates into a local flux impact on the substrate. The selection mechanism is therefore governed by the spectral eigenvalue λ_n .

6.2 DBI Softening as Dynamical Spectral Filtering

In the weakly non-linear regime the discrete Dirac–Born–Infeld dynamics produces an amplitude-dependent frequency

$$\omega_n^2(A_n) \approx \lambda_n \left(1 - \frac{3\lambda_n A_n^2}{8c_\chi^2} \right).$$

This simplified expression follows from the identity

$$\frac{\kappa_n}{\|\psi_n\|_\infty^2} = 1$$

which holds for all non-trivial sectors of the Q_8 Cayley graph.

The non-linear correction therefore depends only on the Laplacian eigenvalue. Modes with larger λ_n undergo stronger softening as the effective amplitude increases.

The relational substrate thus behaves as a dynamical low-pass spectral filter.

6.3 Conceptual Interpretation

The mechanism uncovered in this toy model can be summarised as

substrate capacity $\rightarrow$ spectral admissibility $\rightarrow$ nonlinear filtering $\rightarrow$ surviving sectors.

On the Q_8 Cayley graph the smallest non-trivial eigenvalue

$$\lambda_{\rho_4} = 6$$

belongs to the quaternionic non-abelian sector. This sector therefore possesses the largest admissible amplitude window and is the least affected by DBI softening.

The non-perturbative DBI reduction derived in Section 4.6 shows that the admissibility envelope

$$A_n^{\max} = \frac{c_\chi}{\sqrt{\lambda_n}}$$

is not merely a linear approximation. It coincides with the exact saturation boundary of the one-mode DBI dynamics on the Q_8 Cayley graph.

The bounded relational capacity of the substrate therefore acts not merely as a spectral cut-off but as a dynamical selector, favouring the most resilient spectral sectors.

6.4 Beyond the Q_8 Toy Model

The identity

$$\frac{\kappa_n}{\|\psi_n\|_\infty^2} = 1$$

is a special feature of the small and highly symmetric Q_8 graph.

On larger graphs this ratio will generally depend on the structure of the eigenmodes. The non-linear DBI correction may therefore amplify the spectral selection mechanism rather than acting uniformly across sectors.

7 Structural Origin of the Admissibility Hierarchy and the SU(2) Emergence Thread

7.1 The ratio $\sqrt{4/3}$ is not accidental

The admissibility hierarchy established in Proposition 2,

$$\frac{A_{\rho_4}^{\max}}{A_{\rho_k}^{\max}} = \sqrt{\frac{8}{6}} = \sqrt{\frac{4}{3}} \approx 1.155, \quad (52)$$

has a precise internal origin in the Q_8 Cayley spectrum. It originates entirely from the Laplacian eigenvalues $\lambda_{\rho_4} = 6$ and $\lambda_{\rho_k} = 8$. The ratio $8/6$ is in turn fixed by the adjacency eigenvalues $\mu_{\rho_k} = -2$ and $\mu_{\rho_4} = 0$ through $\lambda = d - \mu$ with $d = 6$.

The central observation is that $\mu_{\rho_4} = 0$. The quaternionic sector lies exactly at the centre of the adjacency spectrum. This is a structural consequence of the character theory of Q_8 , not a numerical coincidence.

7.2 Why $\mu_{\rho_4} = 0$: character-theoretic argument

For any Cayley graph on a finite group G with symmetric generating set S , Schur's lemma gives the adjacency eigenvalue in the irreducible sector ρ as

$$\mu_\rho = \frac{1}{\dim \rho} \sum_{s \in S} \chi_\rho(s). \quad (53)$$

For the quaternionic representation ρ_4 of Q_8 , the generating set $S = \{\pm i, \pm j, \pm k\}$ maps to the matrices

$$\rho_4(\pm i) = \pm \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad \rho_4(\pm j) = \pm \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \rho_4(\pm k) = \pm \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}. \quad (54)$$

All six generator matrices are traceless. Therefore

$$\chi_{\rho_4}(s) = \text{tr}(\rho_4(s)) = 0 \quad \forall s \in S, \quad (55)$$

and the character sum in Eq. (53) vanishes identically, giving $\mu_{\rho_4} = 0$.

This is a structural property. The generators of Q_8 map to traceless matrices under ρ_4 because Q_8 is generated by the unit quaternions i, j, k , whose matrix images are exactly the imaginary units of $\text{SU}(2)$.

7.3 Q_8 as the minimal discrete encoding of spin

The connection between Q_8 and spin is not coincidental but topological. The simply-connected double cover of $SO(3)$ is $\text{SU}(2)$, which is topologically the three-sphere S^3 :

$$\text{SU}(2) \cong \{a_0 I + ia_1 \sigma_x + ia_2 \sigma_y + ia_3 \sigma_z : a_0^2 + a_1^2 + a_2^2 + a_3^2 = 1\} \cong S^3. \quad (56)$$

The group $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ is the smallest non-abelian subgroup of the unit quaternions, hence the smallest non-abelian subgroup of $\text{SU}(2)$. Its elements encode the minimal subgroup structure compatible with spinorial $\text{SU}(2)$ geometry.

7.4 Connection with the S^3 spectral invariant $\lambda_2/\lambda_1 = 8/3$

On the three-sphere S^3 , the Laplacian eigenvalues are $\lambda_\ell = \ell(\ell + 2)$ with multiplicity $(\ell + 1)^2$, giving

$$\ell = 0 : \lambda_0 = 0, \quad \ell = 1 : \lambda_1 = 3, \quad \ell = 2 : \lambda_2 = 8, \quad \ell = 3 : \lambda_3 = 15. \quad (57)$$

The fundamental spectral ratio of S^3 is

$$\left. \frac{\lambda_2}{\lambda_1} \right|_{S^3} = \frac{8}{3}. \quad (58)$$

This ratio is the primary spectral invariant used elsewhere in the Cosmochrony programme to characterise the S^3 geometry from the relational spectrum [?].

The ratio from the present Q_8 calculation,

$$\frac{\lambda_{\rho_k}}{\lambda_{\rho_4}} = \frac{8}{6} = \frac{4}{3}, \quad (59)$$

is related heuristically to the S^3 ratio by

$$\frac{\lambda_{\rho_k}}{\lambda_{\rho_4}} = \frac{1}{2} \cdot \left. \frac{\lambda_2}{\lambda_1} \right|_{S^3}. \quad (60)$$

7.5 The $\text{SU}(2)$ emergence programme

The Q_8 result motivates a systematic programme toward $\text{SU}(2)$ emergence via spectral admissibility. The key chain of inclusions is

$$Q_8 \subset 2I \subset \text{SU}(2), \quad (61)$$

where $2I$ is the binary icosahedral group of order 120.

The asymptotic step is provided by the Ramanujan graphs of Lubotzky, Phillips, and Sarnak [?], which are Cayley graphs of $\text{SL}(2, \mathbb{F}_p)$ designed to approximate S^3 with near-optimal spectral gap.

7.6 Physical interpretation

The non-abelian quaternionic sector has zero net correlation with the generators of the graph, $\mu_{\rho_4} = 0$. This neutrality is the source of its resilience. The bounded-flux constraint imposes less penalty on modes that do not couple positively or negatively to the relational connectivity.

The mechanism is therefore a structural consequence of the embedding $Q_8 \hookrightarrow \text{SU}(2)$ together with bounded substrate capacity:

$$\text{bounded relational capacity} \implies \text{non-abelian sector resilience.} \quad (62)$$

8 Spectral Admissibility on the Binary Icosahedral Group $2I$

8.1 The group $2I$ and its place in the programme

The binary icosahedral group $2I$ is the double cover of the icosahedral rotation group $A_5 \cong \text{PSL}(2, \mathbb{F}_5)$, and has order $|2I| = 120$. It is the largest finite subgroup of $\text{SU}(2)$ and sits in the inclusion chain

$$Q_8 \subset 2I \subset \text{SU}(2), \quad (63)$$

making it the natural next test case for the spectral admissibility programme.

The irreducible representations of $2I$ are labelled by spin and have dimensions matching the low-spin content of $\text{SU}(2)$:

$$\underbrace{1}_{\text{spin } 0}, \quad \underbrace{2, 2'}_{\text{spin } \frac{1}{2}}, \quad \underbrace{3, 3'}_{\text{spin } 1}, \quad \underbrace{4, 4'}_{\text{spin } \frac{3}{2}}, \quad \underbrace{5}_{\text{spin } 2}, \quad \underbrace{6}_{\text{spin } \frac{5}{2}}. \quad (64)$$

Of these nine irreps, six are spin-type in the sense that they arise from restrictions of $\text{SU}(2)$ representations.

8.2 Conjugacy classes and character table

The group $2I$ has nine conjugacy classes, enumerated by element order:

Class	Order	Size	Notation
$1a$	1	1	identity
$2a$	2	1	central element -1
$3a$	3	20	
$4a$	4	30	
$5a$	5	12	
$5b$	5	12	algebraic conjugate of $5a$
$6a$	6	20	
$10a$	10	12	
$10b$	10	12	algebraic conjugate of $10a$

The class sizes satisfy

$$1 + 1 + 20 + 30 + 12 + 12 + 20 + 12 + 12 = 120.$$

Writing $\varphi = (1 + \sqrt{5})/2$ and $\bar{\varphi} = 1/\varphi = \varphi - 1 = (\sqrt{5} - 1)/2$, the character values of the

spin-type irreps are

	$1a$	$2a$	$3a$	$4a$	$5a$	$5b$	$6a$	$10a$	$10b$
χ_1	1	1	1	1	1	1	1	1	1
χ_2	2	-2	-1	0	$\bar{\varphi}$	$-\varphi$	1	φ	$-\bar{\varphi}$
χ_3	3	3	0	-1	φ	$-\bar{\varphi}$	0	$-\bar{\varphi}$	φ
χ_4	4	-4	1	0	-1	-1	-1	1	1
χ_5	5	5	-1	1	0	0	-1	0	0
χ_6	6	-6	0	0	1	1	0	-1	-1

(65)

8.3 Generating set: order-10 elements

A natural generating set for the comparison with Q_8 is the union of the two order-10 conjugacy classes,

$$S = 10a \cup 10b, \quad |S| = 24. \quad (66)$$

This choice is canonical because it is inversion-symmetric, spinorially analogous to the Q_8 generating set, and generates $2I$.

8.4 Adjacency eigenvalues and spectral hierarchy

For a generating set that is a union of conjugacy classes, the adjacency eigenvalue in the irreducible sector ρ is

$$\mu_\rho = \frac{1}{\dim \rho} \sum_{C \subseteq S} |C| \chi_\rho(C). \quad (67)$$

For $S = 10a \cup 10b$, with both classes of size 12, this gives

Sector	μ_ρ	$\lambda_\rho = 24 - \mu_\rho$	$A_\rho^{\max} = c_\chi / \sqrt{\lambda_\rho}$
χ_1 (trivial)	24	0	∞
χ_2 (spin- $\frac{1}{2}$)	6	18	$c_\chi / \sqrt{18}$
χ_3 (spin-1)	4	20	$c_\chi / \sqrt{20}$
χ_4 (spin- $\frac{3}{2}$)	6	18	$c_\chi / \sqrt{18}$
χ_5 (spin-2)	0	24	$c_\chi / \sqrt{24}$
χ_6 (spin- $\frac{5}{2}$)	-4	28	$c_\chi / \sqrt{28}$

(68)

The spectral ordering of the non-trivial spin-type sectors is therefore

$$\lambda_{1/2} = \lambda_{3/2} = 18 < \lambda_1 = 20 < \lambda_2 = 24 < \lambda_{5/2} = 28. \quad (69)$$

8.5 Algebraic origin: the golden ratio identity

For the spin- $\frac{1}{2}$ sector,

$$\mu_2 = \frac{1}{2} (12 \chi_2(10a) + 12 \chi_2(10b)) = 6 (\chi_2(10a) + \chi_2(10b)). \quad (70)$$

Using

$$\chi_2(10a) + \chi_2(10b) = \varphi + (-\bar{\varphi}) = \varphi - \bar{\varphi} = 1, \quad (71)$$

we obtain

$$\mu_2 = 6. \quad (72)$$

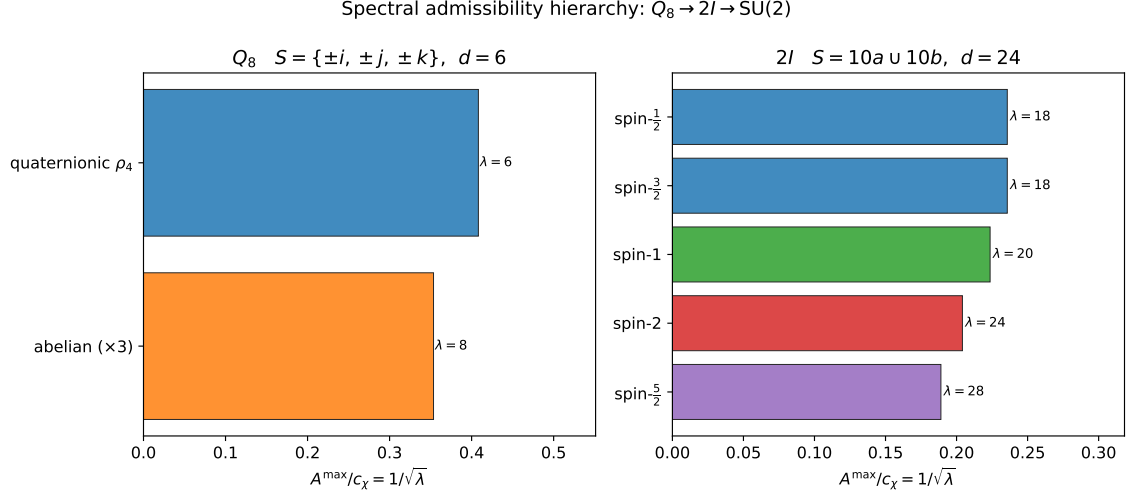


Figure 3: Spectral admissibility hierarchy from Q_8 to $2I$. Left: on Q_8 , the quaternionic sector has the smallest non-trivial Laplacian eigenvalue. Right: on $2I$ with generating set $S = 10a \cup 10b$, the hierarchy resolves into the ordered spin-type sectors $\lambda_{1/2} = \lambda_{3/2} = 18 < \lambda_1 = 20 < \lambda_2 = 24 < \lambda_{5/2} = 28$. The comparison illustrates the sharpening of the admissibility hierarchy along the chain $Q_8 \subset 2I \subset \text{SU}(2)$.

For the spin- $\frac{3}{2}$ sector, both classes have character +1:

$$\chi_4(10a) = \chi_4(10b) = 1, \quad \mu_4 = \frac{12 + 12}{4} = 6. \quad (73)$$

8.6 Comparison with Q_8 and the strengthening hierarchy

On Q_8 with $S = \{\pm i, \pm j, \pm k\}$,

$$\frac{A_{\text{best}}^{\max}}{A_{\text{worst}}^{\max}} = \sqrt{\frac{8}{6}} = \sqrt{\frac{4}{3}} \approx 1.155. \quad (74)$$

On $2I$ with $S = 10a \cup 10b$, the corresponding ratio is

$$\frac{A_{1/2}^{\max}}{A_{5/2}^{\max}} = \sqrt{\frac{28}{18}} = \sqrt{\frac{14}{9}} \approx 1.247. \quad (75)$$

Thus the admissibility contrast increases from Q_8 to $2I$:

$$\sqrt{\frac{4}{3}} \approx 1.155 \quad \longrightarrow \quad \sqrt{\frac{14}{9}} \approx 1.247. \quad (76)$$

8.7 Physical interpretation: towards low-spin selection

On Q_8 , the admissibility hierarchy separated the non-abelian sector from the abelian sectors. On $2I$, the hierarchy is resolved across several spin sectors:

$$\text{spin-}\frac{1}{2}, \text{spin-}\frac{3}{2} \succ \text{spin-1} \succ \text{spin-2} \succ \text{spin-}\frac{5}{2}, \quad (77)$$

where $\succ$ denotes a strictly larger admissibility window.

The ordering is not strictly monotone in spin, since the spin- $\frac{1}{2}$ and spin- $\frac{3}{2}$ sectors are degenerate. Even so, the hierarchy is clearly biased toward lower spins.

9 Next Steps

The programme has now established spectral admissibility on two finite subgroups of $SU(2)$:

Step	Group	Key result
1–3	Q_8	$\lambda_{\text{non-ab}} = 6 < \lambda_{\text{abel}} = 8$; admissibility ratio $\sqrt{4/3}$
4	Q_8	structural origin via Pauli tracelessness ($\mu_{\rho_4} = 0$)
5	$2I$	four-level hierarchy; ratio $\sqrt{14/9}$; origin via $\varphi - 1/\varphi = 1$

Both results are exact, non-perturbative, and derived from character theory alone. The hierarchy strengthens along the inclusion chain $Q_8 \subset 2I \subset SU(2)$, and its algebraic origin is structural at each step.

Step 6: Large Cayley graphs approaching $SU(2)$. The natural continuation is to test whether the spectral hierarchy persists and sharpens along a family of Cayley graphs whose size grows without bound while the geometry converges toward $SU(2) \cong S^3$. The complete programme chain is

$$Q_8 \longrightarrow 2I \longrightarrow \text{SL}(2, \mathbb{F}_p) \xrightarrow{p \rightarrow \infty} SU(2), \quad (78)$$

where the final arrow denotes local spectral convergence of the Cayley graphs toward the S^3 Laplacian.

Natural candidates are the Ramanujan graphs of Lubotzky, Phillips, and Sarnak [?] on $\text{SL}(2, \mathbb{F}_p)$. These graphs have three properties that make them ideal for the programme:

1. *Near-optimal spectral gap.* The non-trivial Laplacian eigenvalues satisfy $\lambda \geq d - 2\sqrt{d-1}$ (Alon–Boppana bound), where $d = p+1 = |S|$. The admissibility envelope $A_\rho^{\max} = c_\chi / \sqrt{\lambda_\rho}$ therefore remains uniformly controlled as p grows.
2. *Increasing size.* $|\text{SL}(2, \mathbb{F}_p)| = p(p^2 - 1)$ provides a sequence of progressively finer discrete approximations to S^3 .
3. *Full spin tower in the limit.* The irreducible representations of $\text{SL}(2, \mathbb{F}_p)$ include all half-integer spins $j = 0, \frac{1}{2}, \dots, \frac{p-1}{2}$, reproducing the complete spin spectrum of $SU(2)$ as $p \rightarrow \infty$.

Step 6 consists of four sub-tasks.

Step 6a (generating set). The LPS construction provides an explicit symmetric set of $p+1$ generators of $\text{SL}(2, \mathbb{F}_p)$, defined by the solutions to $a^2 + b^2 + c^2 + d^2 \equiv 1 \pmod{p}$ with a odd and b, c, d even, embedded via the standard quaternion-to-matrix map. This set plays the same structural role as the generators $\{\pm i, \pm j, \pm k\}$ of Q_8 and the order-10 elements of $2I$.

Step 6b (character computation). For any Cayley graph generated by a union of conjugacy classes, the adjacency eigenvalue in the irreducible sector ρ is

$$\mu_\rho = \frac{1}{\dim \rho} \sum_{C \subseteq S} |C| \chi_\rho(C). \quad (79)$$

The character tables of $\text{SL}(2, \mathbb{F}_p)$ are classical, and the spin- j sector corresponds to the representation $\text{Sym}^{2j}(V_2)$, exactly as in the Q_8 and $2I$ cases.

Step 6c (admissibility envelope). The Laplacian eigenvalue and admissibility window for the spin- j sector are

$$\lambda_j = (p+1) - \mu_j, \quad A_j^{\max} = \frac{c_\chi}{\sqrt{\lambda_j}}. \quad (80)$$

Step 6d ($p \rightarrow \infty$ limit). The key observable is the ratio

$$R_p = \frac{A_{j=1/2}^{\max}}{A_{j=j_{\max}}^{\max}} = \sqrt{\frac{\lambda_{j_{\max}}}{\lambda_{1/2}}}, \quad j_{\max} = \frac{p-1}{2}. \quad (81)$$

The programme predicts $R_p > 1$ for all finite p , meaning that the spin- $\frac{1}{2}$ sector remains the most admissible sector along the sequence.

If the ordering $\lambda_{1/2} < \lambda_1 < \dots < \lambda_{j_{\max}}$ becomes monotone for large p and $\lim_{p \rightarrow \infty} R_p > 1$, this would establish

$$\text{bounded relational flux} \implies \text{monotone low-spin survival} \quad \text{in the SU(2) limit.} \quad (82)$$

The results on Q_8 and $2I$ already provide the first two non-trivial instances of this mechanism. Their algebraic origins — Pauli tracelessness and the golden-ratio identity $\varphi - 1/\varphi = 1$ — are both instances of the same underlying principle: the generating set S is positively correlated with the low-spin representations in character. This yields $\mu_{\text{low spin}} > \mu_{\text{high spin}}$ and therefore $\lambda_{\text{low}} < \lambda_{\text{high}}$.

Whether this structural correlation persists uniformly along the chain (78) is the central open question of Step 6.